

Slide 01

Causora

0:00-0:35 / ENGLISH NARRATION

Causora helps a procurement team review a supplier renewal before committing money. I will explain the algorithm, how the website runs a real AI review, and the design choices that make this project distinctive. The core idea is to make a recommendation inspectable. A reviewer should be able to trace a number to a calculation, trace a contract constraint to evidence, and see whether the AI response belongs to the same simulation. This demonstration uses reviewed synthetic data, so the results illustrate the mechanism rather than a validated forecast for a real company.

After this paragraph, advance to slide 02.

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day3/monte_carlo.py

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/agents/wanghao-day3/agent_day4/pipeline.py

Slide 02

The renewal decision

0:35-1:15 / ENGLISH NARRATION

The case contains a contract interaction that a simple renewal summary can miss. The locked forecast is twenty six thousand units. A sixty percent minimum share commits fifteen thousand six hundred units to Supplier A. In the demand contraction scenario, demand falls fifteen percent to twenty two thousand one hundred units, but the commitment still uses the locked forecast. A rolling sixty percent benchmark would be thirteen thousand two hundred and sixty units. The difference is two thousand three hundred and forty units. That is excess commitment relative to the benchmark, not automatically physical surplus inventory. We compare keeping A, diversifying with B while respecting the minimum, and exiting A with a twenty five thousand dollar fee.

After this paragraph, advance to slide 03.

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Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day1/interfaces.py

Slide 03

Monte Carlo sampling

1:15-2:05 / ENGLISH NARRATION

The simulator runs one thousand trials across one hundred and four weeks. Each trial represents a possible sequence of demand and deliveries. It samples historical weekly demand with replacement and also samples Supplier A lead times from observed records. Supplier B has insufficient history, so the implementation uses an explicitly reviewed triangular distribution. That uncertainty needs disclosure. The generator uses PCG64 and a recorded seed. Crucially, the options share the same master draws. This is the common random numbers method. Differences between options therefore reflect their policies under comparable sampled worlds rather than unrelated lucky draws. Repeating the same inputs and seed supports reproducibility. The bootstrap still relies on historical representativeness and does not model every possible future dependency.

After this paragraph, advance to slide 04.

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day3/monte_carlo.py#L81

Slide 04

Inventory and cost calculation

2:05-2:55 / ENGLISH NARRATION

For each week, the engine receives orders that are due, combines arrivals with opening stock, and fills as much demand as inventory allows. Any remaining demand becomes shortage. It then uses inventory position, including stock already on order, to decide replenishment. Contract commitments and the fixed allocation policy still apply. Across each complete trial, the engine accumulates purchase cost, holding cost, shortage loss, renewal premium, and any exit fee. Expected total cost is the average of those trial costs. P90 cash uses the nearest rank rule and excludes shortage loss from cash spending. Stockout probability means the fraction of trials with any lost units. Service level instead measures filled units divided by demand. Those two measures answer different questions and should never be treated as complements.

After this paragraph, advance to slide 05.

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day3/monte_carlo.py#L100

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day3/monte_carlo.py#L223

Slide 05

Constraint screening before cost ranking

2:55-3:40 / ENGLISH NARRATION

Selection is deterministic. The engine first screens contract feasibility, the risk threshold, and the cash ceiling. It ranks expected total cost only among eligible options. In this recorded baseline run, D1 costs three hundred and forty eight thousand and forty dollars, with a stockout probability of zero point eight percent. D0 and D2 fail the twelve percent risk threshold. D1 therefore survives screening and wins the cost ranking. The AI does not invent the eligible option. The same rule also prevents choosing an apparently cheap option in a stress scenario when it violates the risk limit. If no option survives, the system reports no feasible option instead of forcing a recommendation. These values belong to the reviewed synthetic case.

After this paragraph, advance to slide 06.

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day1/interfaces.py#L193

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/backend/backend/simulation_day3/monte_carlo.py

Project evidence: work/day6/02-live-matrix.png

Slide 06

Real AI review through OpenRouter

3:40-4:35 / ENGLISH NARRATION

After simulation, the backend sends a bounded review through OpenRouter. CFO, COO, and Risk review their assigned perspectives in parallel. A Critic then examines the combined views, including the interaction between falling demand and locked commitments. The preferred Critic uses a different model family, although that alone does not prove independent reasoning. A Synthesizer creates a structured brief. The normal path uses five provider calls. A bounded correction or fallback can add one. The transport records the returned model, provider call identifier, usage, and verified cost. Validators check numeric references, required evidence, and run identity before accepting the brief. Those checks reduce specific errors but cannot guarantee that every qualitative statement is correct. We can demonstrate actual provider calls and structured review, not inspect a model's hidden thought process.

After this paragraph, advance to slide 07.

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/agents/wanghao-day3/agent_day4/pipeline.py#L202

Source: https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/agents/wanghao-day3/agent_day4/openrouter_provider.py#L825

Slide 07

Runtime and human decision control

4:35-5:20 / ENGLISH NARRATION

The public website uses a Vercel frontend and a Render Python API. The backend keeps the API key and stores the shared AI budget in PostgreSQL. Before enabling approval or rejection, the frontend checks that the simulation and Boardroom response belong to the same current run and that validation passed. Saved examples and historical cache remain read only. If the AI review fails, the validated matrix and formula trace remain available, and the user can retry only the Boardroom stage. The screenshot shows a recorded successful review, rather than a promise that every future request will succeed. Human controls record the decision locally in the browser. They do not execute a purchase or change a supplier contract.

After this paragraph, advance to slide 08.

Source:

<https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/frontend/app/page.tsx#L128>

Source:

https://github.com/Ding-hub6029/Causora/blob/1c558f50c400f9fd46acbd363be8bfeaa17cbc3f/Day5/Causora/deployment/render_start.py

Project evidence: work/day6/06-live-brief.png

Slide 08

Differentiation through inspectable decisions

5:20-6:10 / ENGLISH NARRATION

Causora's strongest differentiation is the combination of mechanisms. Shared draws make the comparison repeatable. Formula and evidence links expose the basis of a claim. Assigned AI roles and a Critic create a place to challenge the interpretation. Matching run identity prevents a stale brief from silently becoming the current recommendation. A procurement reviewer can therefore examine both the quantitative tradeoff and the reasoning around it. This is a clear positioning opportunity among AI products, but it is not evidence that Causora outperforms every chat assistant or enterprise tool. The project targets one concrete workflow. Its next advantage would come from proving that this workflow helps real users detect a bad commitment and reach a better supported decision.

After this paragraph, advance to slide 09.

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Slide 09

Evidence and remaining limits

6:10-6:55 / ENGLISH NARRATION

Our evidence includes a complete real AI review with five returned stages and a recorded provider cost of about two cents. The simulator also supports repeated seeded runs. These establish that the pipeline can execute and that its calculations are inspectable. They do not establish broad market superiority. We still need real company data, wider user testing, independent review quality benchmarks, and repeated public runs to assess reliability. The current user evidence is limited to one team participant. The reason this project deserves attention is that it gives a concrete answer to how AI advice connects to calculations and accountable human choice. That is the mechanism we can show today, and the foundation we should test next.

Finish the presentation after this paragraph.

Project evidence: `work/day6/postdeploy-online-verification.json`

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